Spanning Tree Protocol
STP

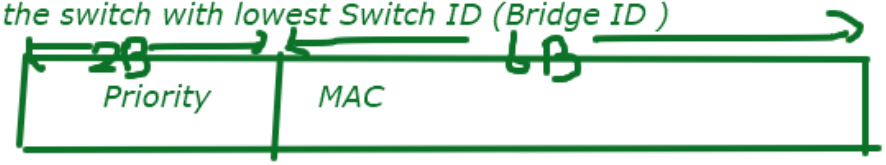
It is L2 protocol which is uesed to prevent switching looping by selection the blocked ports

3e5c.1d2b.1c2f
3e6d.2a1c.2c1b

STP Algorithm

1-Root Switch

It is the switch with lowest Switch ID (Bridge ID)



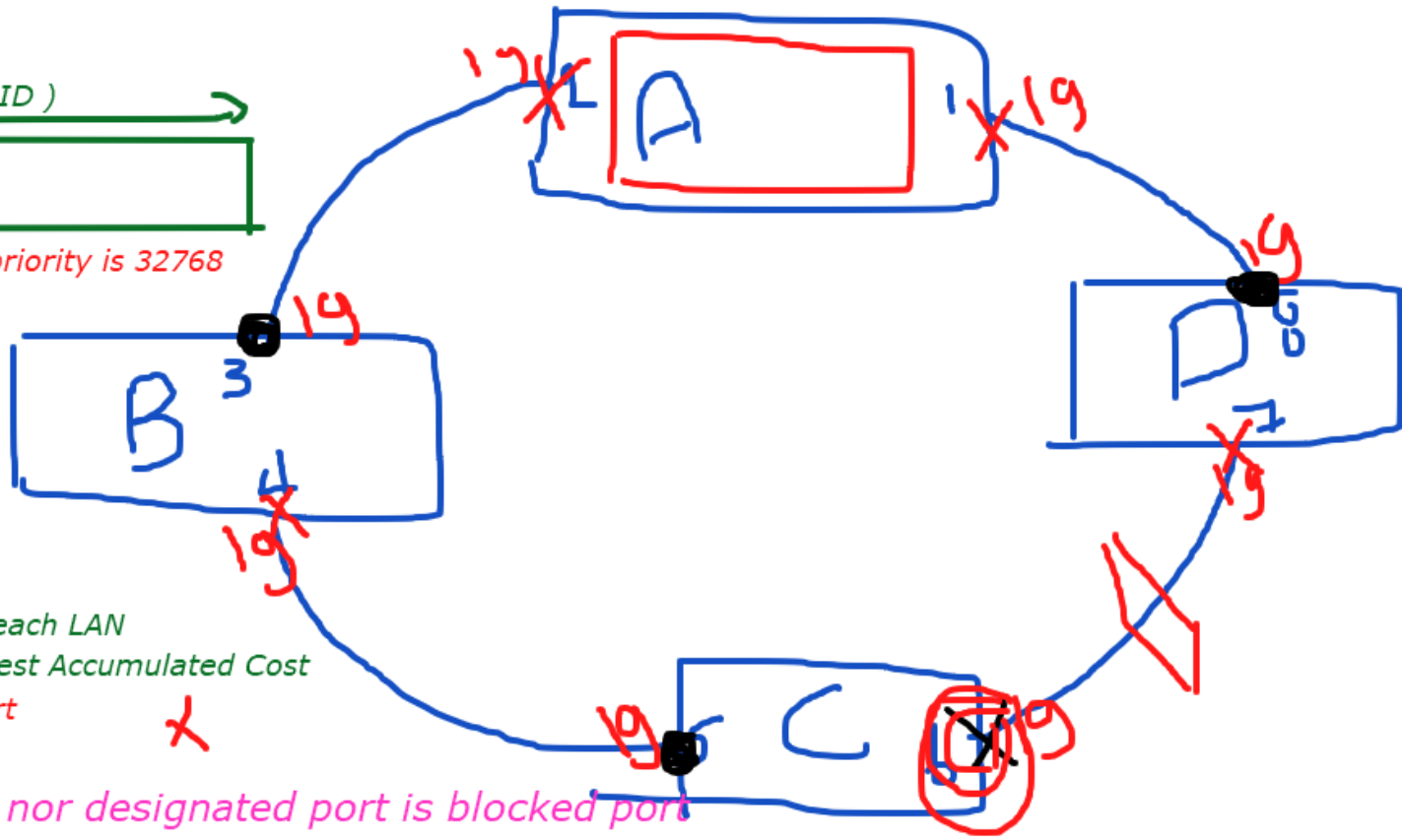
1-Lowest priority
2-Lowest MAC

By default the priority is 32768

2-Root Port

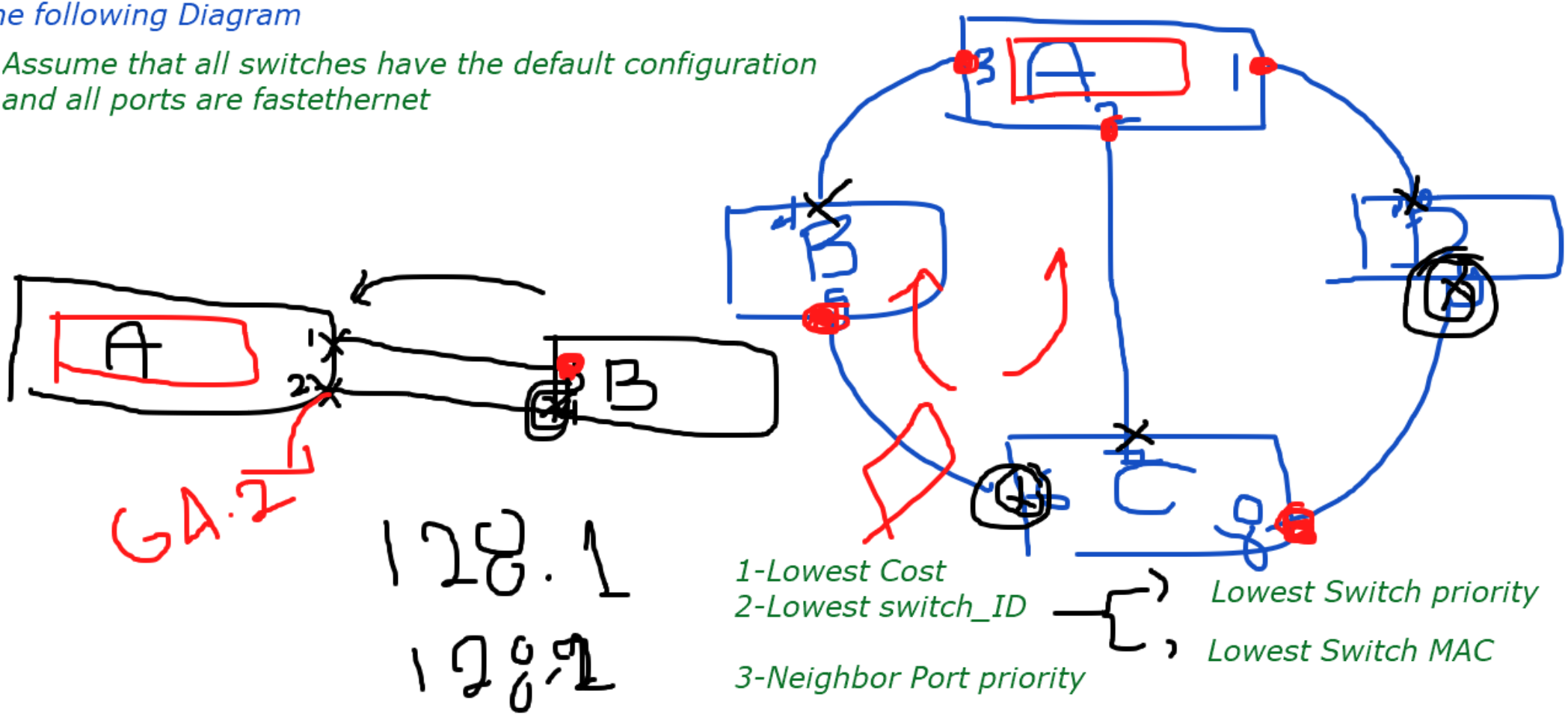
It is only one port per none root switch to reach to root switch with lowest accumulated cost



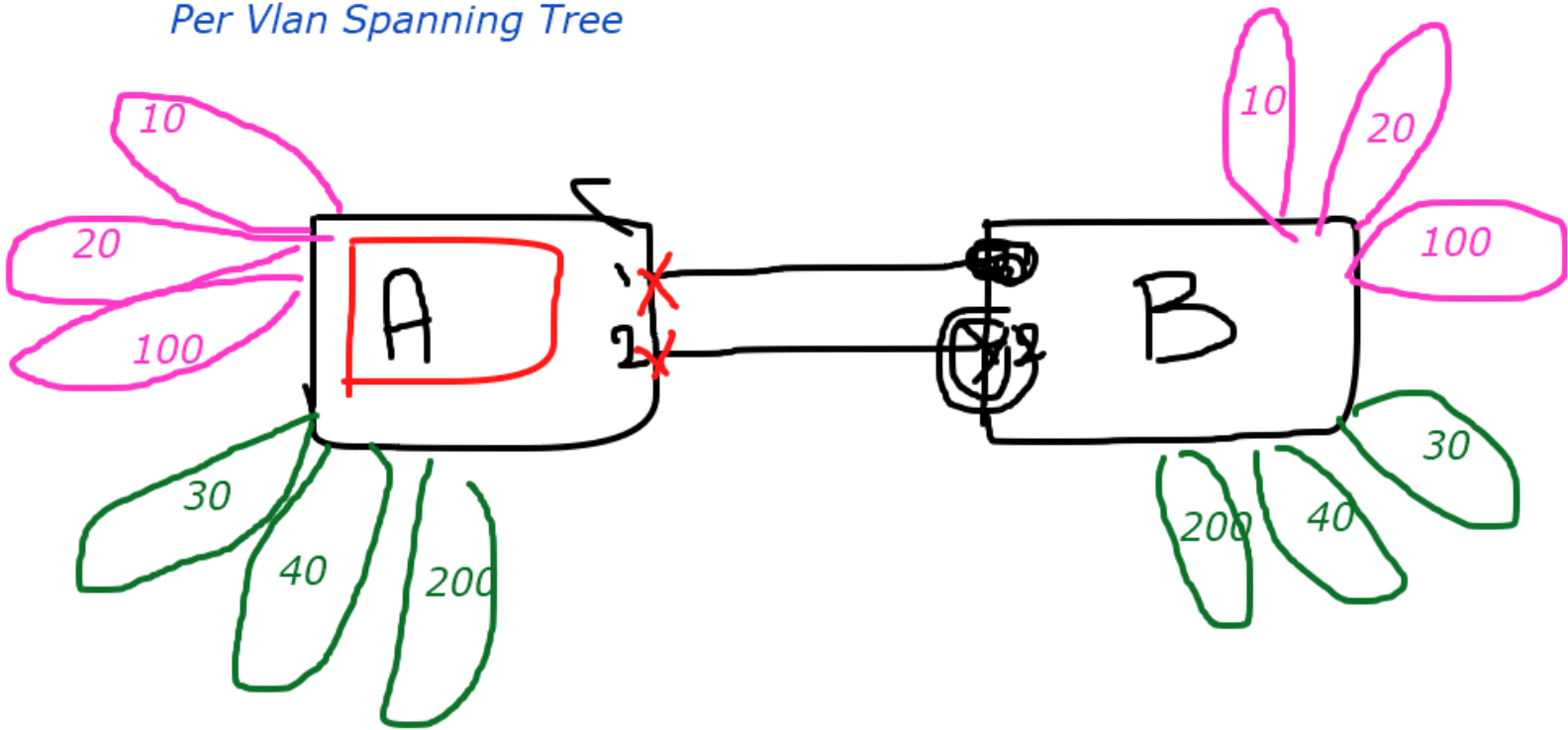

A
B
C
D
19
19
19

Apply the STP Algorithm to elect the blocked port at the following Diagram

Assume that all switches have the default configuration and all ports are fastethernet



PVST
Per Vlan Spanning Tree

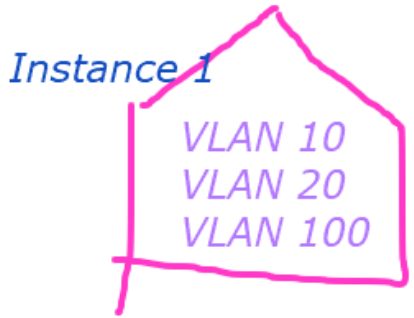


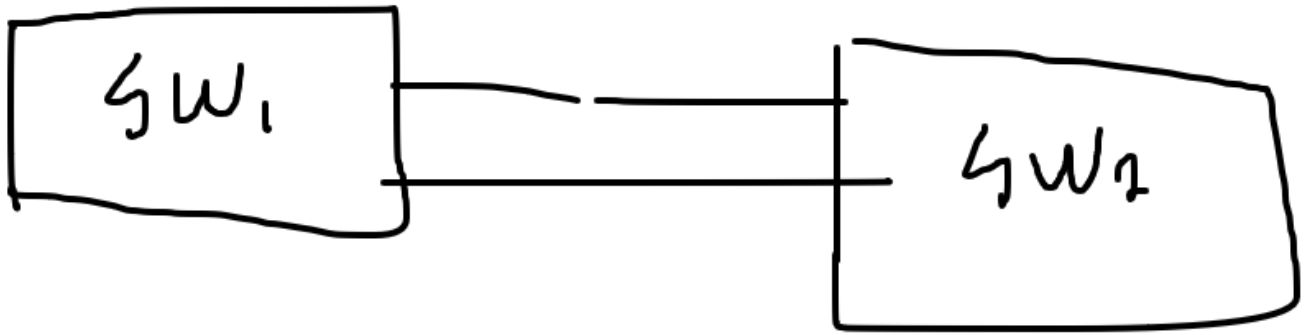
Multiple Spanning Tree

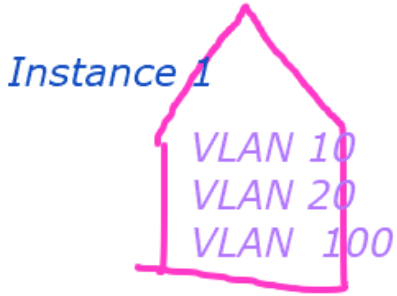
1-It is open standard .

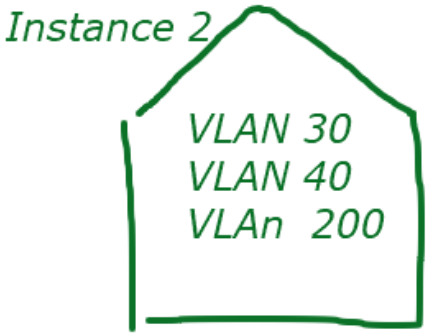
2-It supports the Per vlan STP

3-all Switches must have the same MST name and MST revision











Next Gateway Redundancy Protocol
NGRP

HSRP
Hot standby Router Protocol

VRRP
Virtual Router Redundancy Protocol

GLBP
Gatway load balancing protocol



Hot Standby Router protocol
HSRP

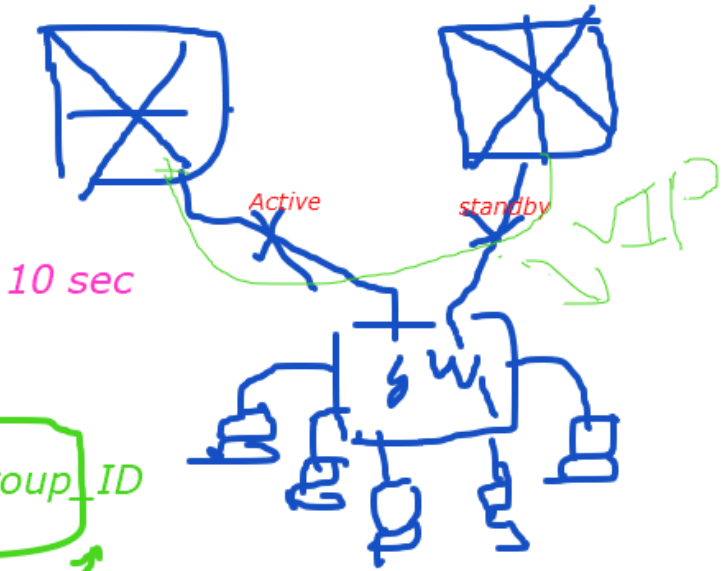
1-It is a cisco protocol

2-It is Created at 1994

3-It support the fault tolerance (Active -passive)

4-HSRP neighbor would send hello packet each 3 sec and holding down is 10 sec

5-HSRP interface has one of two states (Active or Standby)



The MAC address of VIP (Virtual IP) is




```
R1(config)#interface fa0/0
R1(config-if)#standby 5 ip 192.168.33.150
R1(config-if)#standby 5 preemption
R1(config-if)#standby 5 priority  150
```